

Hide and Seek: A Multimodal Approach to Human Detection

Rohan Dalal, Lukas Dimitroff, Akpandu Ekezie, Linnea Funari, Ashley Jang, Jefferson Lin, Shivam Maheshwari, Anshi Paul, Aadhavan Raja Nainar, Srishti Swaminathan, Andrew Xia, Hanning Yan, Joshua Yuen

Abstract

Facial recognition has grown exponentially to become a widely investigated topic in recent years, with human detection acting as the first step in numerous algorithms. Despite these successes, many recognition systems often have trouble in this domain, failing to differentiate between 2D images and actual people. This study specifically focuses on augmenting OpenFace, a facial analysis tool developed in Carnegie Mellon University's MultiComp Laboratory, alongside hardware components to address its shortcomings in human detection. Extensive baseline testing showed that the OpenFace software is poor in delivering accurate and bias-free results, especially with occluded images and long distances. This project aims to tackle these issues by adding sensors (IR Depth, IR Thermal, and mmWave) that integrate motion, human feature, and heat signature detection to OpenFace to be evaluated against the baseline model with a random Forest Classifier. The subsequent model demonstrates a significant improvement in detecting humans accurately when the software is coupled with these external sensing elements, increasing from a 70.37% accuracy before sensor augmentation to a 99.16% accuracy after. Future directions can include more refined testing with non-functional sensors, widening the system's scope of accuracy, and incorporating more components beyond the allotted budget to enhance accuracy.

I. Introduction

A. Historical Background

In the 1960s, a team of researchers gathered to develop a method for computers to recognize faces. Though they did not achieve their goal, their developments in mathematical interpretations of manual markings on the face stand pivotal in contemporary facial recognition software. Fundamentally, current human recognition software still relies on human detection, which is based on the understanding and visualizing certain contours and features of faces that can include facial geometry, expressions, and behavior. By comparing these detected features with other datasets of faces, facial recognition can help determine one's identity. Implementing this software can cost as little as a few thousand dollars. Consequently, these technologies have had profound implications for surveillance and systems. In the last few years, facial recognition has been broadly integrated into devices and software both due to its simplicity and cost-effectiveness. As a result, the market has surpassed \$3.8 billion dollars and is expected to reach \$8.5 billion dollars by 2025.¹

B. OpenFace

Basic recognition software today uses machine learning processes, specifically Neural Networks (NN), to discern human faces. These networks are trained on volumes of facial data propagated through layers of interconnected weighted nodes. One can yield higher accuracy for facial recognition models by tuning their

hyperparameters and finding which set of parameter values produces the best results. Hardware can also influence a model's accuracy: lower quality images may make a human's face look unclear or blurry, making it more difficult for a model to discern whether a human is present. A notable facial detection software is OpenFace, originally designed by Carnegie Mellon University (CMU) students working with the CMU MultiComp Lab. OpenFace is a popular tool used by computer vision and machine learning researchers for facial data augmentation. It uses Deep Neural Networks (DNN) to aid head pose estimation, facial action unit recognition, and eye-gaze estimation.²

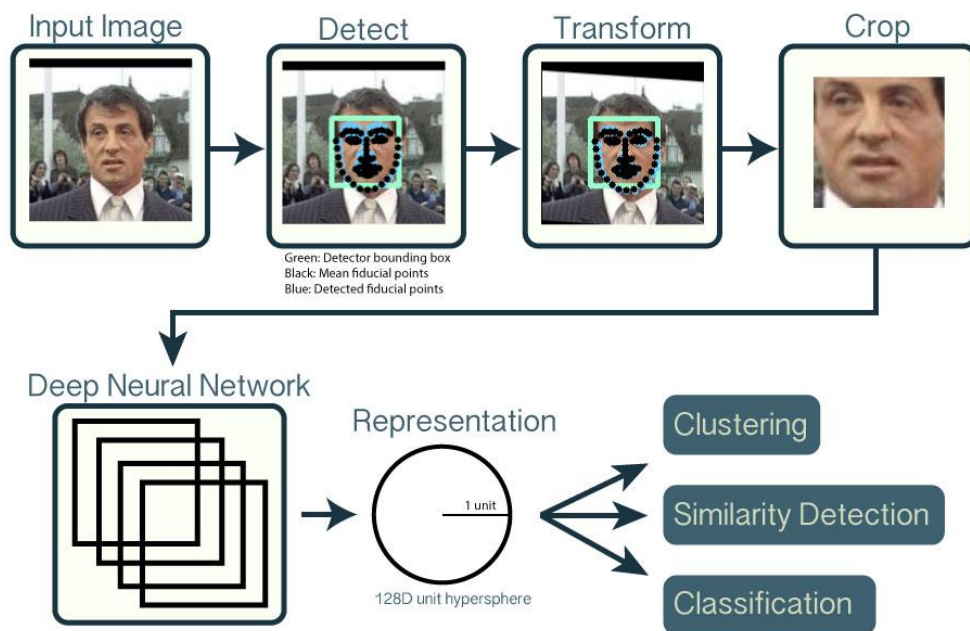


Figure 1: OpenFace schematic³

With an initial input of images, OpenFace's software performs a variety of techniques (detection, transformation, and cropping), feeding it into a DNN for detection. When trained on the unconstrained Labeled Faces in the Wild (LFW) benchmark data, it achieved a state-of-the-art facial detection AUC of 0.973 without hardware assistance.

C. Ethics

Generally, flawed training data can lead large NNs to develop inherent biases. MIT Media Lab's "Gender Studies" analysis, which evaluated commercial gender classifiers, found an error of 34.7% for darker-skinned women but only 0.8% for light-skinned men. If the training data includes mostly individuals with less pigmentation and similar specific characteristics of the face, unsupervised learning becomes critical to reduce bias. Without less biased training sets, facial recognition will always have an inherent gender or racial bias.⁴ Practical usage of facial recognition can greatly harm individuals not well-represented in training data, particularly women of color. Therefore, adopting an equitable approach is necessary for the practical implications of facial recognition.⁵

Beyond internal bias, misuse of human detection and facial recognition software can have terrible implications for civilian privacy and global security. Consequently, it becomes easier for governments to identify people under such pretense for a wide variety of purposes, which nations are already taking advantage of to aid law enforcement agencies or expand surveillance. Due to pricing, along with the growing social and cultural influence of artificial intelligence, the public is growing more conscious of the unintended consequences of this ubiquitous software.⁶

As a result, it is vital to establish clear legal precedents and guidelines for human detection usage. Training data for human detection models should be unbiased towards race and gender to eliminate inherent bias.

D. Related Works

Previous studies have worked on similar applications of improving the recognition from real-life effects of face-covering objects including sunglasses, hats, etc. This effect is pervasive in society and can empirically hinder the accuracy of facial recognition software. Attempts at reducing these errors include changing the software's feature filtering and skin techniques and assisting the DNN from the data cleaning part. These strategies are successful but very case-dependent, and implementing a multimodal approach can often outperform the best facial recognition software, setting an important precedent for future research aiming to refine facial recognition.^{7,8,9,10,11}

In their paper "Face recognition by fusing thermal infrared and visible imagery," researchers George Bebis, Aglika Gyaourova, Saurabh Singh, and Ioannis Pavlidis used thermal imaging for facial recognition specifically to address the facial occlusions caused by glasses. They used genetic algorithms, a search heuristic based on Charles Darwin's Theory of Natural Selection to find the best way to combine visual imagery and thermal imagery. Genetic algorithms work by selecting random individuals from a parent population and choosing them to have children for the next generation: these algorithms are used in a variety of optimization problems. To summarize, other studies have performed similar research, however, we aim to incorporate different types of hardware in combination with a different type of algorithm to create a better way to detect humans before attempting recognition.¹²

E. Goal

Despite its complexity, systems composed solely of facial detection software for human detection purposes can be countered with simple methods like face coverings and crude disguises. We observed that OpenFace will incorrectly recognize the presence of a human being if only a picture of their face is presented. Using external sensors as alternative methods of detection, we aim to augment the usage of OpenFace as human detection by verifying that a human body is present as OpenFace attempts to identify the human's face. The combination of data from the various sensors should be able to make up for the shortcomings of the OpenFace software, especially regarding occlusions, increasing its range and its accuracy in low-light environments.

II. Methodology

A. Initial Thoughts

When deciding what sensors to use, care was taken to choose a diverse range of sensors that also contain a measure of overlap in terms of capabilities. That way, data could be collected independently and then compared with the other sets to gain insight into the strengths and weaknesses of the sensors themselves. Three sensors were chosen: the millimeter wave (mmWave), Infrared (IR) Vision, and IR Depth. These sensors were chosen based on affordability and ease of use. The resulting goal for this investigation was relatively consistent throughout the project to build a sensing array that surpasses OpenFace's capabilities with our baseline model.

B. Baseline Testing

When considering how to best approach our goal, tests were conducted to gauge OpenFace's capabilities and set quantitative benchmarks that could be compared with and without the sensing array. Initially, research was conducted as to the best way to use OpenFace, including testing basic distances with a laptop webcam and later with a higher quality camera better representative of industry standards. The main goal was to test certain occlusions in the human face, such as covering of the eyes, eyebrows, chin, and nose at optimal conditions for OpenFace (i.e. full lighting, good focal length, etc.), and compare this data to that of identical conditions with a full sensing array. To establish a baseline of OpenFace's capabilities with occluded faces, a series of fourteen pictures were taken with different portions of the subjects' faces covered. Sixteen subjects were used, and two sets of data were collected from subjects with glasses, both with and without glasses, to obtain a larger variety of data. This created a total of twenty-one sets. The pictures were taken on a laptop using an external webcam with a resolution of HD 1080p. They were then run through a script, which collected data about the percent certainty.

A secondary goal was solving OpenFace's difficulties with farther distance. With preliminary testing, it was determined that OpenFace loses most of its capabilities around nine feet. To establish a baseline for distance testing, groups of twenty-five pictures were taken at each foot away from the camera from one to twenty-five. Seventeen people had their pictures taken and again, subjects with glasses did two sets, yielding a total of twenty-two collected sets. The pictures were then run through the same script as above.

Despite its successes, OpenFace can easily be fooled through occlusions using simple pieces of paper or small coverings of a person's face, showing how facial occlusions lead to inaccuracies for even the most contemporary facial recognition software. Figure 2 displays an image in which OpenFace was unable to recognize a human, in which the researcher occludes his eyes. The success rate can dramatically decrease in cases where the face's features are occluded. Through testing a set of 21 diverse individuals, it was found that certain parts of face over each other as covering a different feature can entail different responses. Figure 2 displays an image in which OpenFace was unable to recognize a human, in which the researcher occludes his eyes. The success rate can dramatically decrease in cases where the face's features are occluded. Through testing a set of 21 diverse individuals, it was found that the certain parts of face over each other as covering a different feature can entail different responses.²



Figure 2: OpenFace fails to detect occluded image

Features	Successes	Fails	% Successes
top half	3	18	14.29
bottom half	8	13	38.1
right side	11	10	52.38
top right	17	4	80.95
bottom right	19	2	90.48
forehead	21	0	100.0
nose	18	3	85.71
eyes	18	3	85.71
mouth	20	1	95.24
chin	21	0	100.0
middle face	12	9	57.14
diagonal	17	4	80.95
fully covered	0	21	0.0
none covered	21	0	100.0

Figure 3: Occluded faces' success and failure rates

Through this data, conclusions could be drawn based on the frequency of successful detection of certain regions. To do this, the 67 facial landmarks that OpenFace utilizes to aid in its detection model were grouped into major regions: the eyes, nose, eyebrows, jawline, chin, and mouth. Based on the frequency of undetected facial landmarks that were occluded, the frequency of the overall region could be determined

by finding a weighted average. Then, each region's "Occlusion Ratio," or the frequency of undetected occlusions per region, was normalized to ease the comprehension of comparisons.

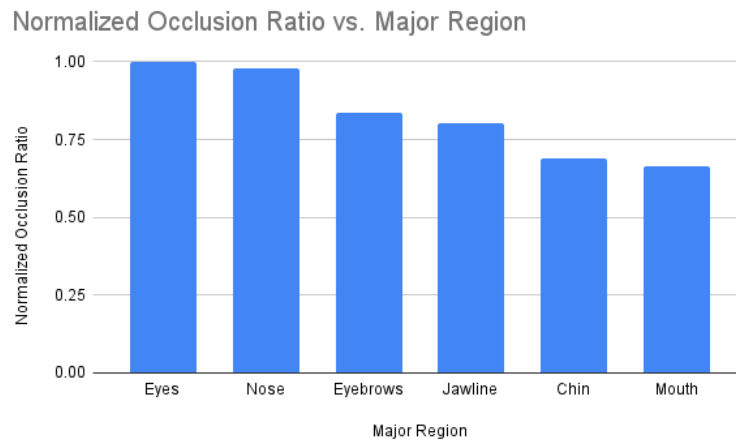


Figure 4: Normalized Occlusion Ratio vs. Major Region

C. Millimeter Wave Sensor

The mmWave is a specific class of radar technology that collects information based on the measurement of electromagnetic waves. Usually composed of Transmit (TX) and radio frequency (RF) components, mmWave sensors have an approximate frequency range of 30-300 GHz to determine the reflected information of the object in the path. This allows for the collection of metadata dependent on the composition of the object, and different substances will have different attributed information. The mmWave's usefulness can be realized in the data collection processes of industry communication and tracking systems such as wearable devices, 5G connections, and vehicle networks.

In recent years, mmWave sensors' ability to sense independently of external light and weather conditions greatly benefited raw facial identification data collection methods. This combination of mmWave and machine learning techniques has had successful real-life applications, encompassing a more diverse range of data visualization. For these reasons, we aimed to include the 24GHz mmWave Sensor (MR24HPC1) Human Static Presence Module Lite into our cost-effective multi-modal facial detection system to achieve remarkable accuracies from the context-specific metadata of mmWave.

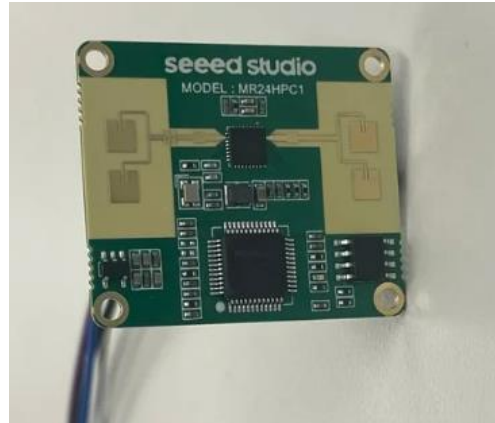


Figure 5: mmWave radar sensor hardware (MR24HPC1)

Specifically, the MR24HPC1 uses hex commands that correspond to the desired task, which in this case should be a combination of motion and human detection. Given initialization hexes, this chosen mmWave radar sensor has a whole host of abilities. Based on that, a receiving hex will be decoded, analyzed, and augmented. From there, we can visualize the data, applying the baseline occlusions to humans. The mmWave sensor is practically useful in both stationary and non-stationary sensing with its powerful ability to still detect humans with various features or coverings.

D. Infrared Vision Sensor (Thermal)

Infrared Vision Sensors, also known as Thermal Cameras, measure the infrared radiation emitting from objects with temperatures greater than 0. Initially used for surveillance and security purposes, their affordability has allowed researchers to use thermal cameras for wide-ranging applications, including human detection. The Adafruit AMG883 is an IR Thermal Camera, adding infrared vision to any Feather mainboards. It contains an 8x8 matrix of thermal sensors, outputting 64 individual pixels of temperature readings. The sensor measures temperatures from 0°C to 80°C. It has an accuracy of $\pm 2.5^{\circ}\text{C}$, can detect up to 7 meters, and has a frame rate of at most 10 Hz.¹³ The thermal sensor detects the difference between the ambient temperature and a person's temperature for human detection and declares a person is there. If a picture of a human face is shown to OpenFace, it recognizes the picture as an actual human; thermal cameras can address this problem by verifying this result through thermal sensing.

The Adafruit AMG883 was connected to the Raspberry Pi using I2C. I2C consists of SDA and SLC wires, which use a serial communication protocol for connection. Initially, a grayscale temperature graph was created and applied to our output data. This created a reasonable visual representation of the data but was not satisfactory due to the low resolution. The 8x8 output matrix was then expanded through bicubic interpolation into a 32x32 matrix of thermal readings, allowing higher definition and a clearer gradient. To get a better visualization, Matplotlib allowed us to output the 32x32 matrix of thermal readings as a colormap; one example of an output is shown in Figure 7.¹⁴ However, the heatmap is unable to capture real-time data of events. To get data with timestamps, a different approach is necessary.

By averaging out all the individual data points at any given time, the current average temperature can be found. Because timestamps are recorded, the average temperature at different time frames can be stored. Using these values, the increase in temperature at different timestamps can be measured. Later, two different modes of operation were implemented to fill two different roles.

The standard mode actively adjusts how many values it stores by deleting the oldest values and adding new ones to create a constantly changing temperature average. This method is ideal for places with variable temperatures, but the major shortcoming is that it actively incorporates heat signatures into the environment over time. If a person stands in the frame for long enough, they will cease to be recognized.

In response, we created a second mode. In this mode, the sensor takes input data for a set amount of time, and then stops actively updating this list. Thus, every heat signature sensed afterwards is still registered. Although this limits the scope of the sensor, it does not affect any sensor placed in a stable environment, such as rooms maintained at a similar temperature or generally placed inside. We decided to use this method for our experiments, as they take place indoors.

To facilitate data collection, we detected room temperature and calibrated the sensor by recording 400 frames without humans in the frame and averaging them to form a singular room temperature. This calibration is done every run of the program and represents the average temperature in the room. From there, we were able to find the deviation of an incoming temperature from that value and applied a sigmoid curve (Figure 6) to get a percent certainty from the real time temperature values. After fine-tuning multiple constants to better represent the data at hand, our graph could detect heat signatures to around 5 feet with reasonable confidence.

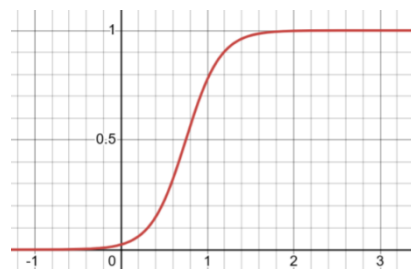


Figure 6: Sigmoid function

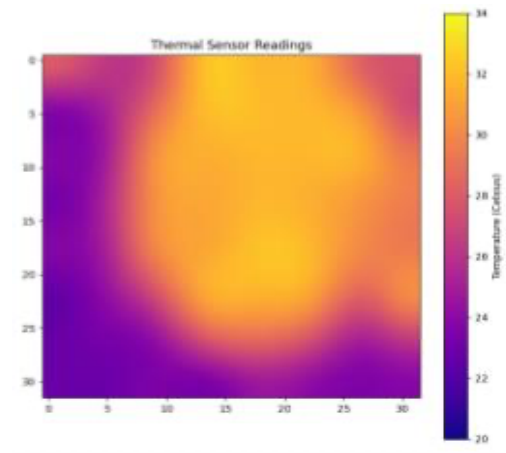


Figure 7: Heatmap

E. Infrared Depth Sensor

We utilized an IR depth sensor, the Microsoft Kinect, for its ability to track depth in three dimensions with infrared (IR) light. The Kinect contains three essential components: an RGB video camera, a multi-array microphone, and an IR depth sensor. For our research, we focused on using the RGB video camera and the IR depth sensor. The RGB camera detects the color components, body type, and facial features with its 640x480 pixel resolution.¹⁵



Figure 8: IR depth sensor (Kinect v1 1414)

Meanwhile, the IR depth sensor contains a light-capturing monochrome sensor and infrared projector to produce a 3D image of the sensor's environment. The infrared projector transmits invisible infrared light and measures the amount of time it takes for the light to reflect off each object. The returning infrared signals are eventually detected by the monochrome sensor; then using the infrared light "flight time distances" to create an accurate scene depth map, the Kinect can infer body position through machine learning.¹⁵

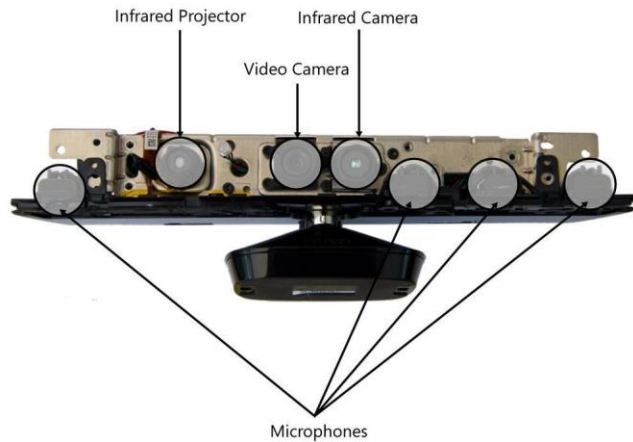


Figure 9: Diagram of Kinect parts¹⁶

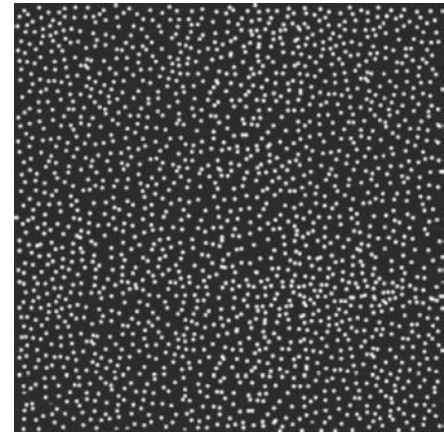


Figure 10: Speckled map¹⁷

The speckled map (Figure 10) is the pattern of IR light that the Kinect sends out and allows the device to evaluate depth with more accuracy. The depth map is created by combining structured light (analyzing a speckled pattern of infrared laser light) with computer vision techniques: depth from focus and depth from stereo. Depth from focus estimates depth by inspecting several images from the same perspective. Objects further away are less focused, so decreased quality indicates a further distance away from the sensor. On the other hand, depth from stereo incorporates different viewpoints of the same image to identify depth, mimicking the way human eyes perceive depth.¹⁵



Skeleton Overlay on Test Depth Maps - Linnea & Anshi

Figure 11: Skeleton Overlay from the Kinect Explorer WPF Showcasing Joint Inferencing

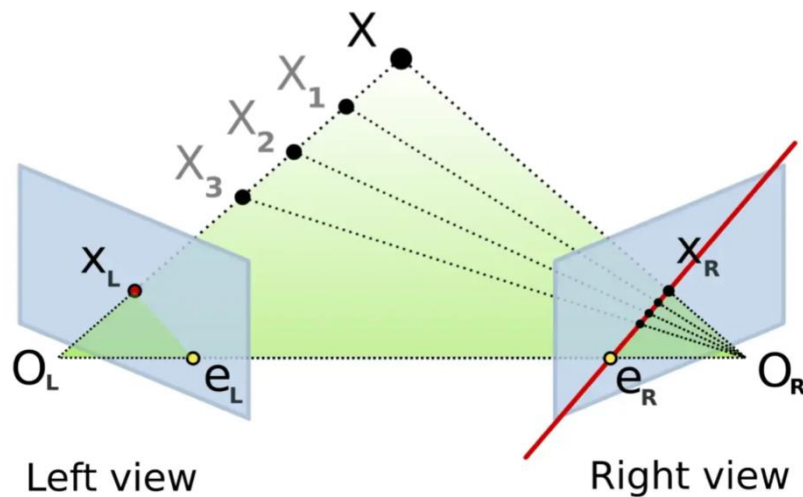


Figure 12: Schematic of Depth Mapping Technique¹⁸

Initially, Microsoft intended to recognize humans with the Kinect by tracking body movements. Over time, scientists turned to tracking and labeling 20 different joints on a coordinate system, as displayed in Figure 13. The Kinect team trained a random forest model to best identify the 20 different joints (see the appendix for more about random forests).¹⁵

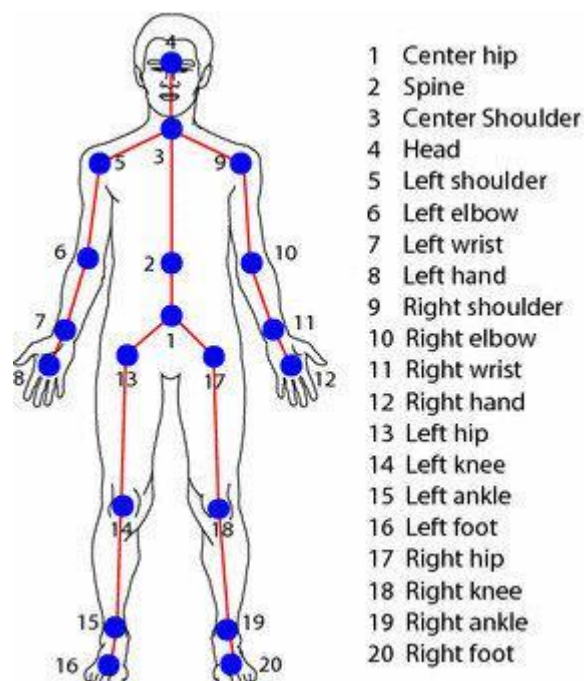


Figure 13: 20 distinct joints detected by Kinect v1¹⁹

As we used the older Kinect v1, the latest software development kit (SDK) v2.0 could not be used nor was it applicable to the capabilities of the older version of the Kinect. The SDK v1.8 was made for the Kinect v1;

however, the documentation for this SDK is either limited or not accessible, so we discovered much of the information regarding SDK use through educated guess and check methods. For example, the joint values were determined through piecing together the puzzle of known joint values and guessing the surrounding joint values. As opposed to the Kinect v2 that collects a multitude of additional joints for skeletal analysis and tracking, the Kinect v1 records 20 distinct points to craft skeletons as previously mentioned.

To address the occlusion limitations of OpenFace specifically, we created a way to record data in a set amount of time from the Kinect into a readable format. The data collected in CSV (comma separated values) format included the timestamp of each frame, the skeleton tracking ID, joint coordinates in the x, y, and z direction for all 20 joints, and the confidence levels for every joint. The tracking ID was used to determine if there was interference in the recording. For example, if someone else walked into the frame, there would be multiple tracking IDs identified. Alternatively, if the Kinect detects something nonhuman to be a skeleton in the same frame, a “ghost,” we can also identify the ghost in the recorded data. The confidence values for every joint from the Kinect were determined by their tracking state, which returns whether a joint is being tracked or is inferred. In the Kinect Toolkit Browser’s Kinect Explorer WPF application, the inferred skeleton limbs were represented by thin lines, which appeared when the subject was either out of frame or occluded. If the limb was actively being tracked, a thick line was drawn in between those two points. This was quantified by assigning a confidence level of 1 if the joint is being tracked; a confidence level of 0.5 if the joint is being inferred; and a level of 0 if no joint is detected. In the C# application written, if the Kinect does not find a skeleton, there is no information written to the csv. Thus, we can assume a confidence level of 0 if there is no information in the csv file.

The officially supported languages for the SDK include C++, Visual Basic, and C#. However, we can use other languages to access the Kinect sensor using the library OpenKinect which supports a host of other languages including Python. For simplicity, we used C# and the SDK to create the script to communicate with the Kinect. We created a .NET Console Application that included `Microsoft.Kinect` as a reference in `KinectSkeletonRecorder.csproj` to access the tools available in the SDK. When run with `dotnet run`, the application first asks the user what to name the csv. During data collection, the files were labeled in the following naming convention: `[number]_[initials]_[occlusion type].csv`. For example, a recording of Anshi Paul with a full-face occlusion would be labeled `2_ap_whole_face.csv` and would be placed in a folder titled Anshi. The numbering system keeps each occlusion ordered correctly for data analysis, and the application then asks for the recording duration. After receiving confirmation, the Kinect begins recording for the set amount of time. The application does not record frames in which the Kinect does not detect a skeleton rather than returning null.

To collect occlusion data, we ran the same occlusions as the OpenFace baseline testing and thermal testing. To adapt for the Kinect requirements of “ideal conditions,” we recorded subjects at 7.6 feet to view their entire skeleton with some tolerance for movement. Due to the motion profiling characteristics of the recording algorithm, we also recorded each subject for 5 seconds per occlusion. The Kinect has trouble recognizing occluded people if that is all that is presented to it from the beginning. To combat this, we take advantage of the video capabilities of the Kinect. For every recording of the Kinect data, a new instance is created for the Kinect sensor. If the subject walks into the occluded state, the Kinect adapts for the shift in joint positions in the head region and still returns a human skeleton. Each subject started at 5 feet for the Kinect to get a baseline human skeleton for every instance, then moved to 7.6 feet and performed the

occlusion over the course of 5 seconds. Between every 5 recordings, we referenced the Kinect Explorer WPF to determine if the subject was still in frame and to check if the sensor was still working properly.

Due to time constraints, a smaller representative sample of people were recorded fully with this setup. In this representative sample, there were two African Americans (one male and one female), one South Asian female, one Caucasian male, and two East Asians (one male and one female). Because IR depth sensors detect the skeletons of test subjects, we believed gender would not play a significant role in identifying whether a test subject was human or nonhuman. However, IR can bounce off different skin tones in varying ways, so we selected a representative sample covering all skin tones for data collection (See Discussion). To match the amount of data points collected by the thermal camera and OpenFace, we identified 4 frames from every subject's recording for data analysis.

The data in the csv files are difficult to visualize, so we also used two visualization tools. Using Matplotlib, a Python library used for graphs and visualizations, a 2D and 3D visualization animation could be created for every file. Although the visualizers were used for data analysis, they were also used throughout the data collection process to confirm if subjects were being picked up in the expected manner.

This initial data was then analyzed with another Python script to compare confidence values (See Discussions).

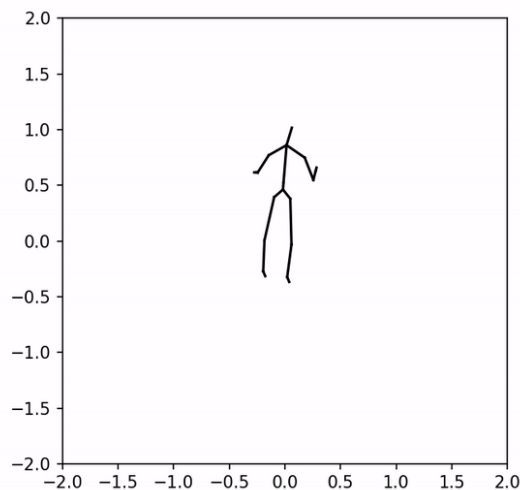


Figure 14: 2D Visualization of Kinect Data

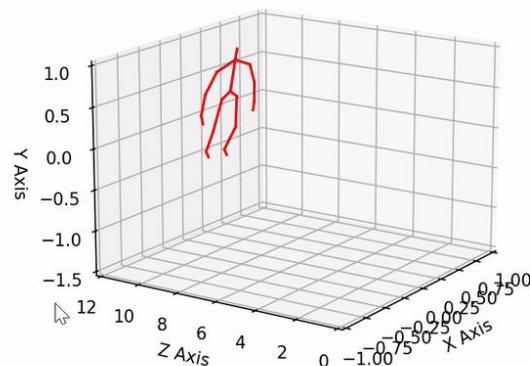


Figure 15: 3D Visualization of Kinect Data

III. Results

A. Infrared Distance

The Kinect data exported was in the form of a csv file, where each row represented a frame of data. The columns include data for 24 joints, with coordinates in three dimensions the confidence level of each joint. A confidence level of 1 means the joint is tracked, a confidence level of 0.5 means the joint is inferred, and

a confidence level of 0 means that the joint is not being tracked. A score of 0.5 is given because, if the skeleton exists, the underlying random forest model used by the Kinect indicates that a human was inherently detected even though the joint was being inferred rather than actively tracked. Therefore, a confidence level of between 0 and 1 accounts for that nuance. The joint coordinates are then plotted in a 3D space as shown in Figure 13, where the facial joint data still exists though it is occluded, but the confidence levels are 0.5 instead of 1. Then, the joints are connected via a line to create “bones.” This data combined creates a visual “skeleton” of data from the detected person or object. Each csv collected can be visualized in a custom 3D visualization tool.

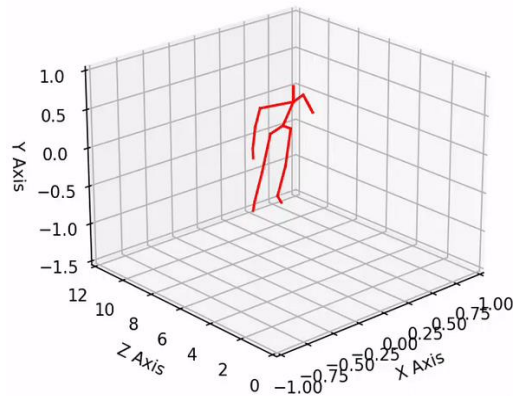


Figure 16: 3D Visualization of Occluded Whole Face from Kinect

The csv files were analyzed with another Python script that is discussed further in Methods and Discussions. From this confidence analysis, the following figures could be generated.

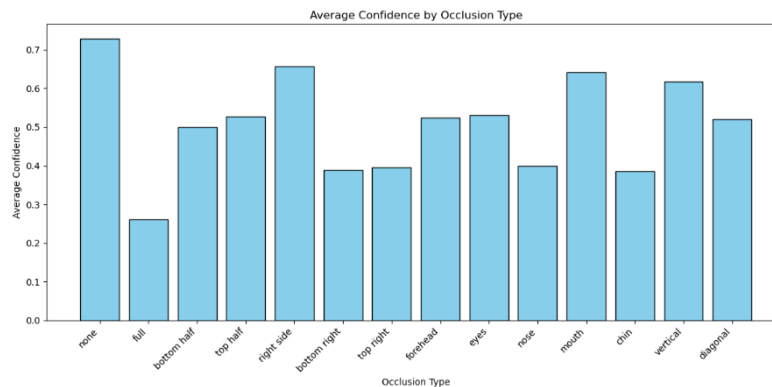


Figure 17: Average Confidence by Occlusion Type Bar Graph from Kinect

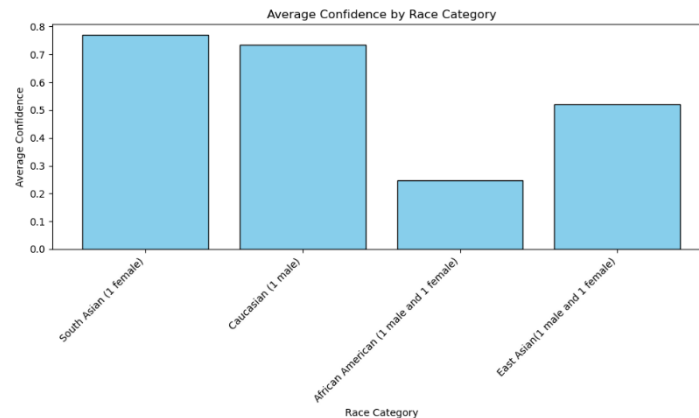


Figure 18: Average Confidence Across All Occlusions by Race Category Bar Graph from Kinect

B. Infrared Vision Sensor (Thermal)

We created a script to detect a change in heat between a space with and without a human present. Each test subject stood 1 foot away from the sensor, and the 14 full and partial occlusions of different facial landmarks were conducted. We found the percentage certainty to directly compare input and output data from the sensor with that of OpenFace.

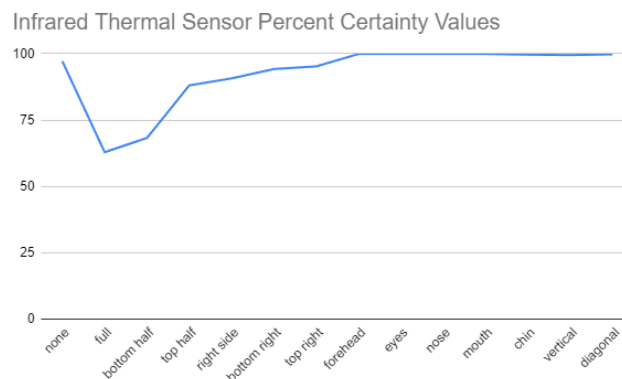


Figure 19: Infrared Thermal Sensor Percent Certainty Values

In Figure 19, each vertex of the graph line represents the average of 15 data points at a certain level of occlusion. It was observed that the main areas experiencing a decrease in certainty were those where the heat signature was dampened: full face covered, half of the face covered, and a quarter of the face covered.

C. Random Forest Model

Finally, we developed a machine learning model to combine all data for prediction. After comparing the receiver operating characteristic (ROC) curves of various algorithms—random forests, support vector

machines, logistic regression, and k-nearest neighbors—we chose to continue our analysis with random forests since it had the highest area under the ROC (AUROC). The ROC curves display the trade-off between sensitivity and specificity. Classifiers that give curves pulled closer to the top left corner indicate a better model performance. The ROC curve for the random forest showed a larger AUROC (Figure 19), which is a general measure of predictive accuracy, and the higher, the more accurate the model would be.

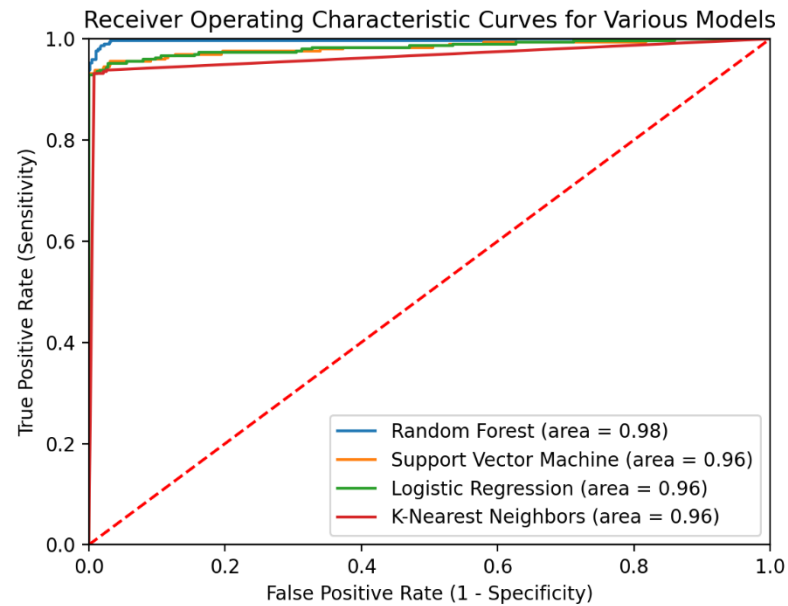


Figure 20: Receiver Operating Characteristic Curves for Various Models

Essentially, our model trains on the facial confidence outputs from OpenFace, the IR heat sensor's average temperature, and IR depth sensors average joint confidence, and is testing using k-fold cross validation. For more details about the random forest model, see the Appendix in section VII.

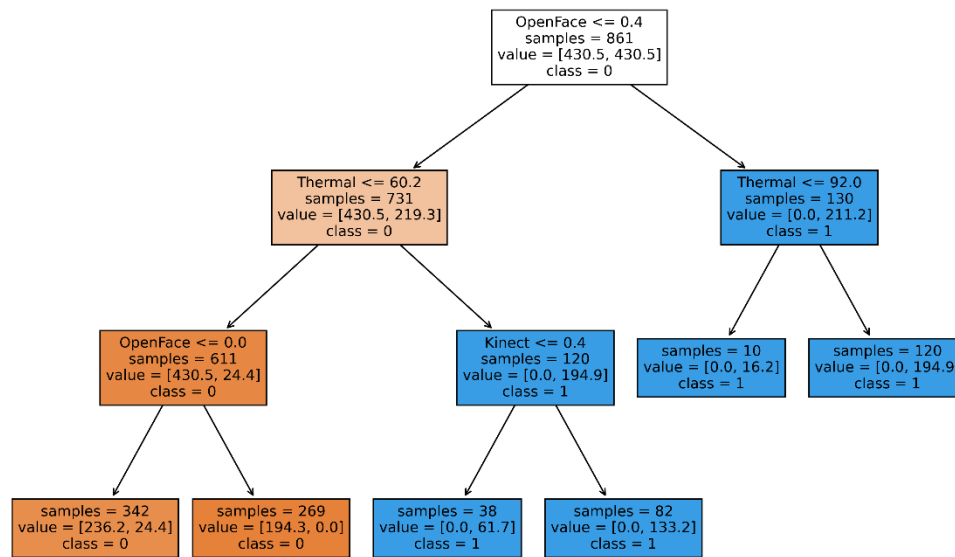


Figure 21: Example of a Random Forest Classifier

The random forest model was then tested using 10-fold cross-validation. This yielded an accuracy of 99.2%, a sensitivity of 97.6%, and a specificity of 99.8%. (Figure 20) The sensitivity represents our accuracy when identifying humans, while the specificity is a measure of how accurately we were able to identify non-humans.

Confusion Matrix Heatmap for the Random Forest Model Predictions

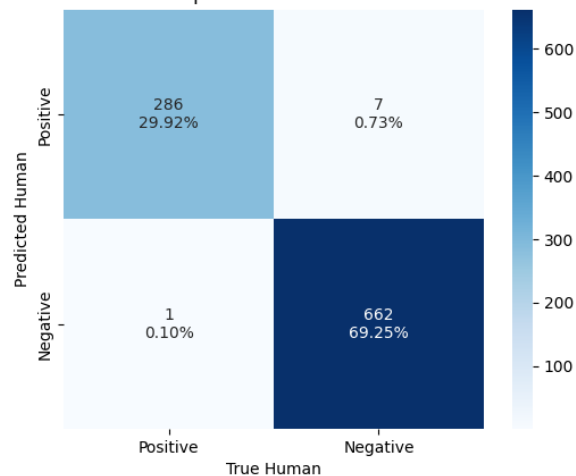


Figure 22: Confusion Matrix of Random Forest Model Results

IV. Discussions

A. Open Face Data Analysis

When testing baseline capabilities of OpenFace, the confidence levels at each foot remained high until around 18 feet (Figure 23), where it could be concluded that OpenFace's recognition capabilities stopped abruptly.

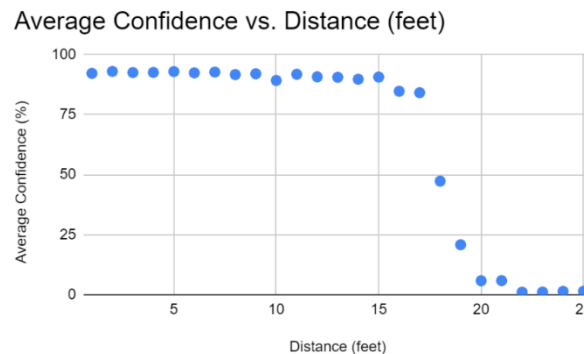


Figure 23: Average Confidence vs Distance from Camera (feet)

Through the occlusion tests done with OpenFace, conclusions could be drawn based on the frequency of successful detection of certain regions. To do this, the 67 facial landmarks that OpenFace utilizes to aid in its detection model were grouped into major regions: the eyes, nose eyebrows, jawline, chin, and mouth. Based on the frequency of undetected facial landmarks that were occluded, the frequency of the overall region could be determined by finding a weighted average. Then, each region's "Occlusion Ratio," or the frequency of undetected occlusions per region, was normalized to ease the comprehension of comparisons.

Figure 4 (See Results) graphs the relative importance of each feature to OpenFace's detection model. For instance, occluding the eyes seems to have more failed detections than any of the other features. With the extensive number of devices that can cover these regions of the face, it would be very difficult to gather and train OpenFace on existing and upcoming occluding devices. This conclusion is what led us to account for other human features to work in conjunction with OpenFace.

B. IR Vision Sensor (Thermal)

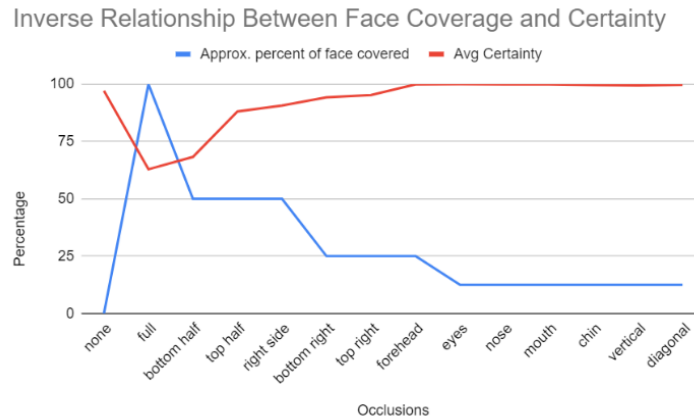


Figure 24: Inverse Relationship Between Face Coverage and Certainty

From the thermal sensor data, we observed that the type of features covered does not largely affect the accuracy of the thermal sensor. Rather, the percentage of the face covered, and thus the percent heat masked, is largely more important, as shown in Figure 24: there is a clear downtrend in certainty when the percentage coverage increases.

The IR Vision data alone can only determine the presence of heat, and not necessarily human presence. But, taken together with the IR Distance data, the sensors complement each other: the distance sensor can detect a human-like structure while the thermal sensor can detect a heat signature. Higher resolution thermal cameras could perhaps be able to distinguish human versus nonhuman heat signatures (See Other Directions).

C. IR Depth Sensor

To utilize the IR Depth data in the overall model, we created a Python script to analyze and isolate the confidence values of every joint recorded. If the Kinect doesn't detect a person's skeleton due to an occlusion, the C# script does not record any data and the recording is cut short. Because the occlusion portion of each recording was in the last 2 seconds of the 5-second recording, if a CSV was spread over a range of less than 3 seconds, we could assume that no occluded skeleton was detected. Using the timestamps recorded, we determined a 3 second recording to be around 75 frames. Therefore, if a CSV file were less than 75 frames, the Python analysis script would automatically give it an average confidence of 0. For every person, we isolated the confidence values (either 0, 0.5, or 1) for each occlusion. Then, we took the average confidence for the joints per occlusion and inputted the confidence levels into a new dataset using a Python script. We repeated this process for every occlusion for all 6 subjects. To increase the number of data points available, we also included 3 additional frames from the 5-second recordings of every subject. However, the skeletons were relatively stable after being occluded, and all the data seemed to be identical to the first data frame. With this, we obtained average confidence values for every occlusion and test subject (with $\frac{3}{4}$ of the subject information being the result of the data augmentation).

Referring to Figure 16, we can observe that the Kinect can adjust for occlusions of the head joint. In this case, we can see a distortion of the head joint; however, the overall skeleton stays intact, so we can still detect a human presence with a facial occlusion. Human presence can still theoretically be detected if clothing occludes some joints, as seen in Figure 11.

After analysis, it became clear that different occlusions had different impacts on the Kinect's ability to retain and adjust the skeleton (Figure 17). For example, the occlusion in which we covered our entire face blocked the entire head node. As seen in visualization playbacks of the recordings, the successes included inferred head data from the Kinect when the folder was occluding the face. In the failed recordings (where a skeleton was no longer identified), the Kinect could not identify a head node, and therefore, the background machine learning model could not detect a human-shaped skeleton, returning a shortened CSV file and an overall confidence rating of 0.

We also noticed that the Kinect still had problems with detecting darker skin tones over lighter skin tones in occluded settings (Figure 18). For example, the average confidence score for Black subjects was significantly lower than the scores for Caucasian and South Asian subjects. As this finding was within a very small sample size of only 6 people, with a larger sample size including more varied skin tones, we could iron out the discrepancies. However, this could be because darker skin tones may absorb more IR waves than lighter ones, reflecting fewer waves to the sensor and decreasing detectability. We could also run the Kinect in combination with our thermal sensor, which would detect a human signature regardless of skin tone.

D. Bias, Error Analysis, and Limitations

In this project, errors in sensor placement could have affected our data. We did not standardize the height between all three sensors during measurement. Some sensors could have been placed in a more optimal position than others, which could have introduced error into our data. Additionally, due to the non-standardized height of participants, many were forced to crouch slightly to present their face at the height necessary to record data. Some participants may not have been directly aligned with the optimal focus area of the sensors. This includes the IR thermal camera, OpenFace, and the IR depth sensor.

Since the data set was small, any potential errors or bias in subjects likely severely affected the model. These potential errors could have come in the form of small differences in clothing and temperature, or large demographic differences in race or age. To build upon our project further, we would include more test subjects: having large datasets would greatly improve this research.

For our sensing array, our thermal sensor only outputs the temperature values of 64 pixels in an 8x8 matrix, which we upscaled into a 32x32 grid using bicubic interpolation. Additionally, as our own testing proves, our thermal sensor could only confidently detect human faces up to a range of 3ft. A higher-quality thermal sensor would have allowed us to collect higher-quality data from the thermal sensor, improving our model.

Another limitation came in the form of the mmWave sensor. It could not function properly due to a technical failure in one rig, and inconsistent data reporting in the other rig. When debugging, the status of the corresponding hex command for human detection in the MR24HPC1 manual would be sent but stuck on receive. Proposed fixes can include but are not limited to new hardware entirely or updating the firmware.

Despite these struggles, the code associated with it can function for other non-human and motion specific hexes.

Along with this, the IR depth sensor has reduced ability to detect objects from a far distance and short distance (we discovered a range of around 4 ft to 10 ft for accurate skeleton detection), cannot record the movement of an object for less than 0.5 seconds, and documents the movement of fewer joints compared to other sensors. Due to time constraints, we were unable to acquire a second-generation Xbox Kinect sensor, which would have increased the resolution of the IR camera, expanded the depth of the sensor, and increased the number of skeleton joints. Future projects would benefit from such a sensor, or even something closer to industry standards such as the Intel RealSense D435 IR depth sensor.^{21, 22}

V. Conclusion

A. Research Summary

To determine the limitations of Facial Recognition technology, we tested OpenFace by occluding certain portions of the face to examine which parts of the face are the most important for OpenFace to make its determination of whether a human is present. To address problems in OpenFace when one's face is occluded, we implemented an IR vision sensor, Xbox Kinect SDK v1.8, and IR thermal sensor, the Adafruit AMG883 IR Thermal Camera FeatherWing. We implemented software algorithms to determine the probability that a human is being detected with each sensor. We tested the same occlusions as OpenFace with the IR vision sensor and IR thermal sensor. Finally, we fed that data into a random forest classifier to determine if a human was in the frame of reference. This led to an accuracy of 99.2% compared to OpenFace's sole occluded face accuracy of 70.37%.

B. Practical Applications

Human detection has a wide range of practical applications across various sectors. Our research aims to contribute to fields such as security, energy conservation, and autonomous vehicles. In security, our model can enhance the surveillance of restricted areas by accurately detecting human presence and promptly alerting the relevant authorities, thereby improving response times and reducing unauthorized access. In airport security, passengers have noticed an increase in facial recognition technology during TSA security checks. Since human detection and facial detection is the first step to facial recognition, multiple aspects of this project could be applied to advance security measures.

In the realm of energy conservation, the model can optimize building management systems by adjusting lighting and HVAC settings based on occupancy, leading to more efficient energy use. Additionally, in the development of self-driving cars, our human detection technology can improve safety by better recognizing pedestrians and other road users, allowing the vehicle to react appropriately to ensure the safety of all involved.

Further, human detection can be used to conserve energy in houses by dimming lights and turning off air conditioning when it detects no humans in the home. Finally, human detection is instrumental in self-driving cars, which need to be able to quickly identify human presence to avoid potentially fatal accidents in

pedestrian areas such as crosswalks and parking lots. In our baseline testing we determined that OpenFace struggles with facial occlusions, especially those surrounding the eye area. Many people wear glasses, masks, and potentially VR headsets in the future, so accurate human detection despite these occlusions is integral to creating safe self-driving cars. Even outside of the realm of self-driving cars, human detection could be applied to driver assist technology such as emergency stops.

We aim for this model to serve as a proof of concept that can be expanded for numerous useful applications to benefit commercial and private industries. With proper expansion and the proper upgrades to sensors, human detection can be expanded much further utilizing a multimodal approach. By improving the accuracy and reliability of human detection, our research has the potential to drive advancements and enhance operations in critical sectors. As detecting humans is the first step in recognizing them, this work can also push human recognition and its applications.

C. Other Directions

Implementing the mmWave sensor could improve our results by implementing a new data source into our model: heartbeat detection. Researchers have discovered that each human has a distinct heartbeat, which could potentially be better than other more common identifiers, such as fingerprints. Given humans have specific heartbeats, measuring them could potentially distinguish human beings from other types of animals, further enhancing our model.²³

With the thermal sensor, edge detection between adjacent pixels remains an unexplored but theoretically viable option. Edge detection would allow us to separate pixels with thermal values of cooler temperatures. This method could potentially outline the picture of a face or another body part, which would help classify whether the test subject in front of the thermal camera is a human or not.



Figure 25: Edge detection²⁰

Implementing motion profiling would have also helped our model by detecting if the object in front of the sensor is moving in a human-like fashion. While human-specific motion profiling was initially the goal of the Kinect use, we decided to use a binary output that depends on the presence of a skeleton or not due to time constraints. Motion profiling works by determining an object's location, velocity, and acceleration. Using

the Kinect sensor, we could train a model to detect the motion of humans versus the motion of inanimate objects and other living beings. Due to the unique reflectivity characteristics of IR waves, the Kinect also tended to see “ghosts,” where a skeleton would be identified when there wasn’t an actual human in frame. This behavior was seen when chairs were in frame. By collecting recordings of actual humans and “ghosts,” another classification model could be trained to further improve the accuracy of human detection by reducing false positives. We have already begun exploration of generating “ghosts” artificially by adding noise to human skeletons: the skeleton still retains a vaguely human shape, the same way a ghost would, yet is distinctly not human in its motion and overall characteristics.

In terms of software, we can further explore models that involve sensor-sensor communication. This would allow for better accuracy in more variable conditions with large amounts of noise. For example, there is a possibility of calibrating the sensitivity of the infrared depth sensor using the infrared vision sensor to allow for better accuracy with regards to identifying ambient heat, or weighting sensors differently depending on detected distance.

Additionally, while our baseline testing also included distance testing to determine the limitations of OpenFace, the focus of this project was simply a proof of concept of the efficacy of a multimodal system to augment OpenFace’s facial/human detection with a focus on occlusions. In the future, the distance realm can be explored further with the use of higher resolution and higher quality sensors with longer ranges.

Another idea we considered but was put away was intersensory communication. How it was planned is that we would have each sensor use information from the others to determine the importance of its reading. For example, if the IR depth told us the object was 8m away, it would accordingly adjust the weight of the thermal since its effectiveness is much less at this range. This approach could help remove many false negatives and false positives, and it would have been implemented given more time.

To improve our model in a different sense, the collection of more data and more training time would mean that we could potentially train and run more complex models, including deep learning models. Using our thermal sensor and Xbox Kinect, we would collect data on humans, inanimate objects, and nonhuman living beings. We would assign each picture to a classification as human or nonhuman and feed the data into a machine learning model of our choice: a few choices for models include logistic regression models and neural networks. With this improved multimodal system, we hope to make strides in human detection and its applications to various fields, whether it be national security, energy conservation, or even self-driving vehicles.

D. Acknowledgements

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E. Code Availability

MmWave Code:

<https://github.com/jeffelin/HumanDet-mmWave>,

Kinect Recording Code:

<https://github.com/li-pearl/kinectv1-skeleton-recorder>

Machine Learning Code:

<https://github.com/shivmaheshwari/HideAndSeek>,

Infrared Vision (Thermal) Code:

<https://github.com/li-pearl/pgss-thermal>

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Appendix A: Random Forest Algorithm

A random forest $R(X)$ is a classifier consisting of a collection D of decision tree classifiers where $|D| = B$ and each tree $d_b(X) \in D$, $b \in \{1, 2, 3, \dots, B\}$ casts a single vote to determine the most popular class for a particular input observation X . To train the algorithm, the bootstrap aggregating, or bagging, technique is used. Given a training set $X = (X_1, X_2, \dots, X_N)$ with corresponding classes $Y = (Y_1, Y_2, \dots, Y_N)$, bagging B times selects B random samples of size M with replacement from the training set and fits decision trees to these samples. To construct these trees, the algorithm builds a maximum tree (splitting each sample up to the last feature) using K randomly selected features and the Gini splitting rule. In this project, we use $K = \lfloor \sqrt{p} \rfloor$, where p is the number of features in each training observation X , because it has been mathematically proven to optimize random forest performance.

When creating decision trees, we want to achieve maximum homogeneity in child nodes each time the data must be divided into two parts, where t_p , t_l , and t_r are parent, left child, and right child nodes; x_i is a feature variable; and x_i^R is the best splitting value for that feature. Doing so optimizes the performance of the random forest.

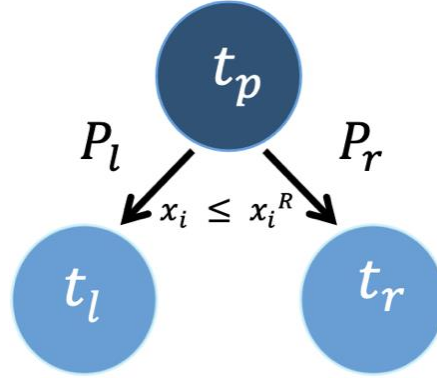


Figure 25: Optimal Split from Parent Node

Figure 25 shows the optimal split from parent node t_p to child nodes t_l and t_r using feature x_i and split value of x_i^R .

To maximize the homogeneity of the child nodes, we use an impurity function $i(t)$. Since the impurity of the parent node t_p is constant for all possible splits $x_i \leq x_i^R, i = 1, 2, 3, \dots, K$, maximizing the homogeneity of the child nodes is equivalent to maximizing the change of the impurity function $\Delta(t) = (t_p) - P_l(t_l) - P_r(t_r)$, where P_l is the proportion of data going to the left child node and P_r the right child node. Therefore, when partitioning the data at each node of a decision tree, the algorithm solves the following maximization problem:

$$[(t_p) - P_l(t_l) - P_r(t_r)]$$

In words, the algorithm will search through all possible values of all features in training set X for the best split question $x_i \leq x_i^R$ that will maximize the homogeneity of the left and right child nodes. In this project, we use the Gini splitting rule, which uses the following impurity function:

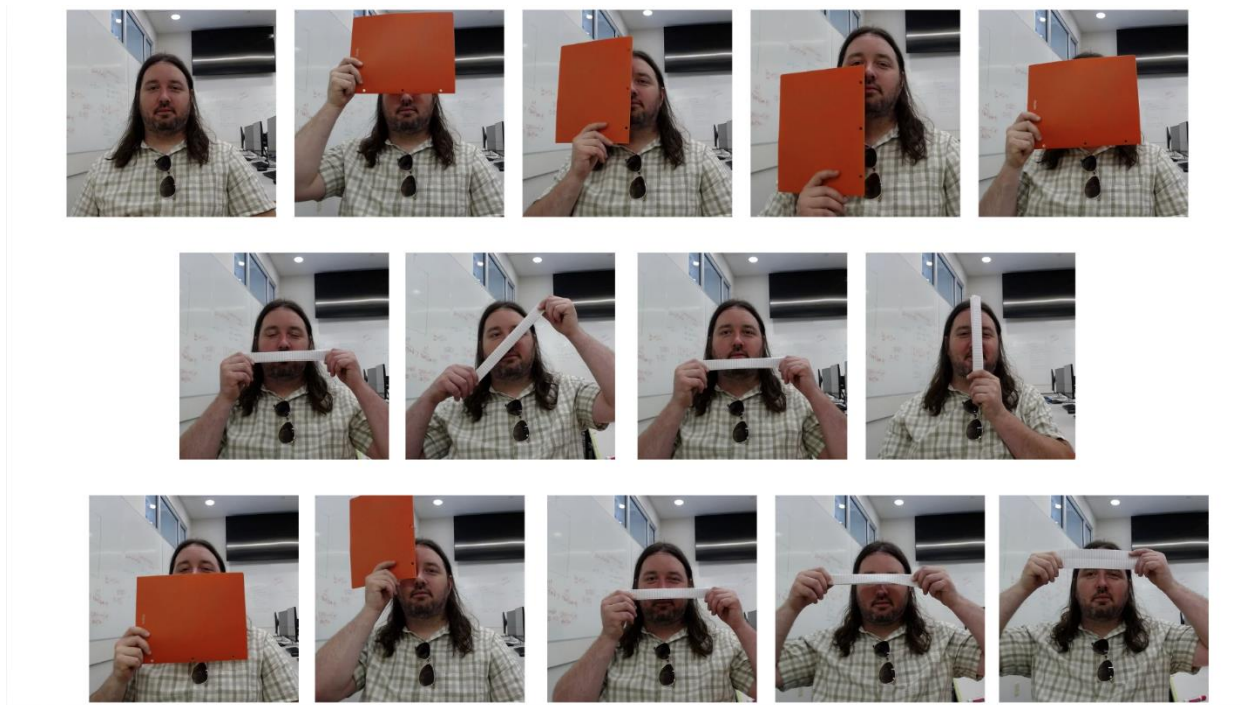
$$i(t) = \sum_{a \neq b} p(a|t) \cdot p(b|t)$$

where a, b and are the indexes of classes $1, 2, 3, \dots, C$, and $p(a|t)$ is the conditional probability of class a provided we are in node t .

After training the algorithm and building the ensemble of decision trees, we can predict classes for a new observation X' by taking the mode of the predictions from all individual decision trees on X' :

$$\hat{R}(X') = \text{mode}(\{\hat{d}_b(X') \mid b \in 1, 2, 3, \dots, B\})$$

Appendix B: Occlusions



The above picture displays the occlusion tests we used for the thermal sensor, IR sensor, and baseline OpenFace testing. The first occlusion is the whole face. The second occlusion covers the top half of the face. The third occlusion covers the left half of the face. The fourth occlusion covers the bottom left of the face. The fifth occlusion covers the entire face. The sixth occlusion covers the mouth. The seventh occlusion covers the diagonal portion of the face, in a way that covers the mouth and right eye. The eighth occlusion covers the chin. The ninth occlusion covers the face vertically in a way that covers the nose. The tenth occlusion covers the bottom half of the face. The eleventh occlusion covers the top left of the face. The twelfth occlusion covers the nose. The thirteenth occlusion covers the eyes. The fourteenth occlusion covers the forehead and eyebrows.